Temporary & Emergency Speed Restrictions

TSR & ESR

Revision Guide | Rule Book Module SP (GERT8000-SP)

Contents

PART 1 — Temporary Speed Restrictions (TSR)

1. Normal Arrangements (SP §3.2)

The following equipment is provided for a standard TSR, in order from the driver's approach:

Equipment	Distance	Notes
Portable AWS magnet	Normally 180 m (≈200 yds) before warning board	May be reduced to min. 45 m (≈50 yds) in some cases
Warning board	Braking distance before speed indicator	Shows the restriction speed; triggers AWS
Speed indicator	At start of restriction	Marks where the TSR begins
[Speed restriction zone]	—	Section of line with restricted speed
Termination indicator (T)	At end of restriction	Alternatively, an Acceleration indicator (A) may be used

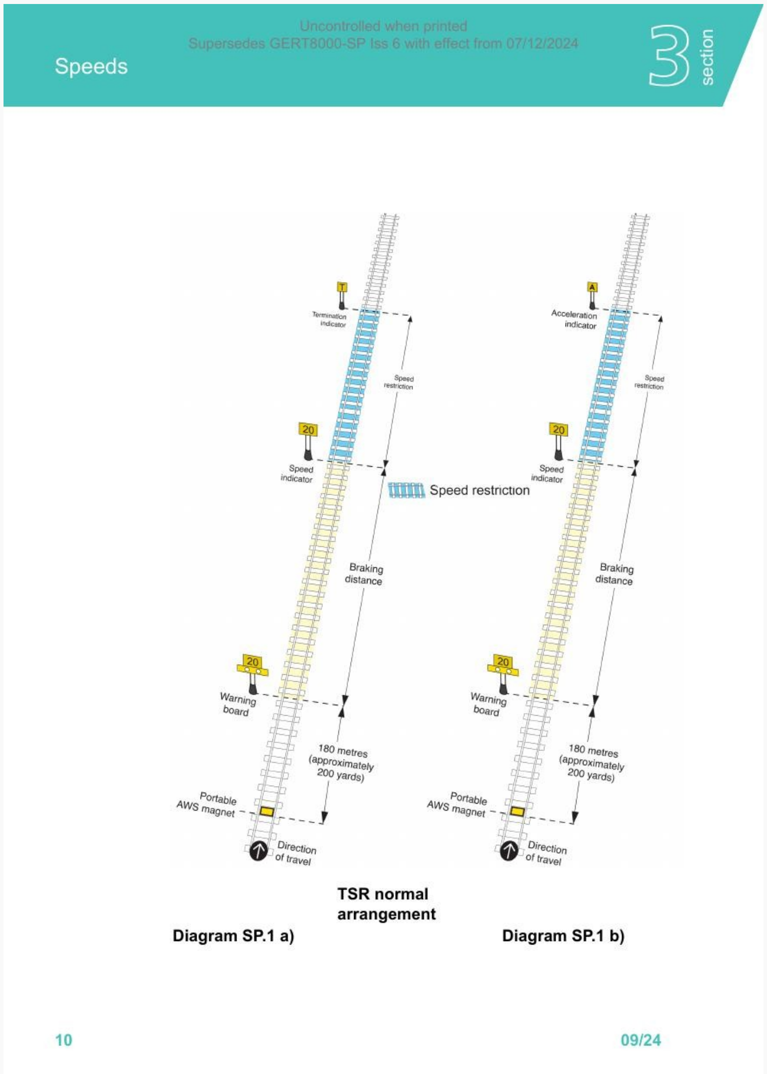


Diagram SP.1 a) Normal TSR | SP.1 b) With Acceleration Indicator — GERT8000-SP

2. Driving Over a TSR (SP §3.1)

- Control your train speed to no more than the speed shown on the warning board.
- Do not increase speed until the whole of your train has passed clear of the lower-speed section.
- Where differential speeds are shown on the warning board, observe the speed that applies to your train type.

> Key point: Speed must be observed from the warning board — not from the speed indicator. You should already be at or below the restriction speed as you pass the speed indicator.

3. Where There is a Fixed AWS Magnet (SP §3.4)

This applies where a fixed AWS magnet already exists near the TSR location, associated with a signal, PSI, or level crossing warning board.

- The warning board must NOT be placed between the fixed AWS magnet and the equipment it applies to.

- If possible, the portable AWS magnet and warning board are kept at the normal distance apart (180 m).
- The distance may be reduced to not less than 45 m (≈50 yds).
- If the warning board is placed at the signal: the associated electro-magnet is disconnected and no portable AWS magnet is provided — but the driver will always receive an AWS warning regardless of the signal aspect shown.

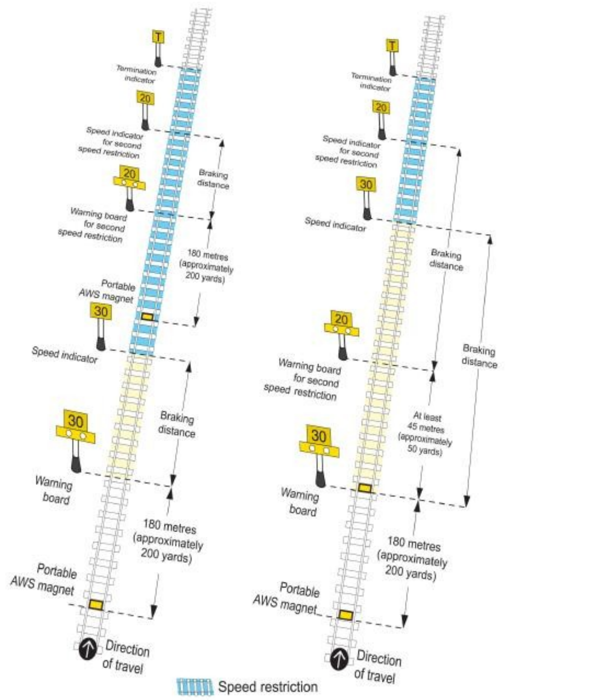
4. TSRs on Single and Bi-Directional Lines (SP §3.5)

- Equipment is provided in both directions of travel.
- A cancelling indicator is normally placed 180 m (≈200 yds) beyond the AWS magnet at each end, facing trains that have already passed through the restriction.
- This distance may be reduced to not less than 45 m (≈50 yds).

5. Consecutive TSRs (SP §3.6)

Where two TSRs sit back-to-back, the board/indicator arrangements depend on the relationship between the two speeds.

Scenario	What changes	Key rule
Lower speed follows higher (e.g. 30 mph then 20 mph) Diagram SP.4 a)	No termination indicator at end of 1st (higher) TSR. Instead, a speed indicator showing the 2nd (lower) speed is placed there.	One termination indicator only — at end of 2nd TSR.
Higher speed follows lower (e.g. 20 mph then 30 mph) Diagram SP.5	No warning board for 2nd (higher) TSR. No termination indicator at end of 1st (lower) TSR. Instead, a speed indicator showing the higher speed is placed there.	One termination indicator only — at end of 2nd TSR.
Not enough distance between boards Diagram SP.4 b)	2nd warning board placed at least 45 m (≈50 yds) beyond 1st warning board. 2nd portable AWS magnet placed immediately beyond 1st warning board.	Minimum 45 m gap between warning boards.



Consecutive TSRs

Diagram SP.4 a) **Diagram SP.4 b)**

Diagram SP.4 a) Consecutive TSRs (lower beyond higher) | SP.4 b) Insufficient distance — GERT8000-SP

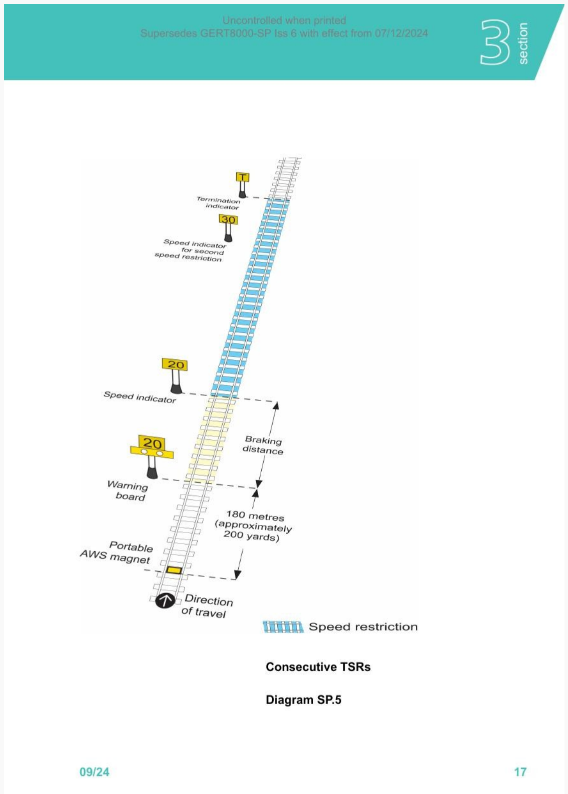


Diagram SP.5 — Consecutive TSRs (higher beyond lower) — GERT8000-SP

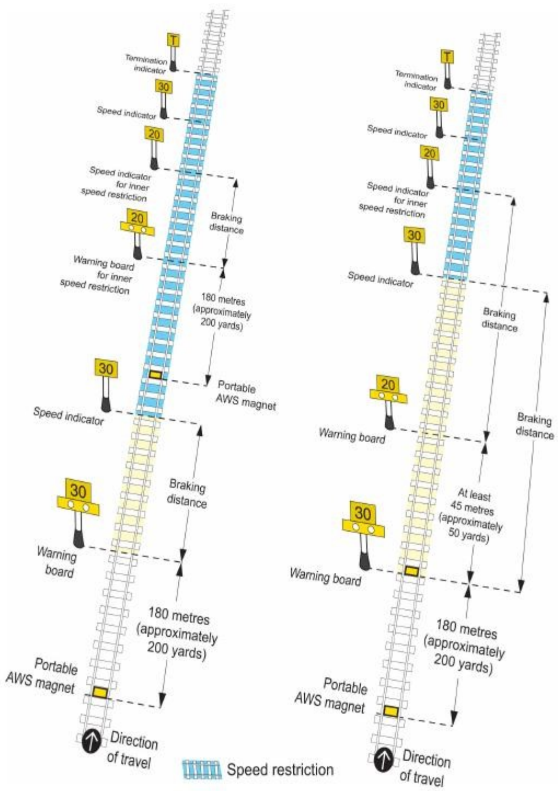
6. One TSR Inside Another (SP §3.7)

Where a second TSR sits entirely within an outer TSR, the arrangements follow the same logic as consecutive TSRs.

Scenario	What changes	Key rule
Lower speed inside higher outer TSR (e.g. outer 30 mph, inner 20 mph) Diagram SP.6 a)	Equipment normal throughout, EXCEPT: no termination indicator at end of inner (lower) TSR. Instead, a speed indicator showing the outer (higher) speed re-appears there.	One termination indicator only — at end of outer TSR.
Higher speed inside lower outer TSR (e.g. outer 20 mph, inner 30 mph) Diagram SP.7	No warning board for inner (higher) TSR. No termination indicator at end of outer (lower) TSR entry. Speed indicator shows inner speed where outer restriction would have terminated; then a further speed indicator showing outer speed re-appears at end of inner	One termination indicator only — at end of outer TSR.

Scenario	What changes	Key rule
	restriction.	
Not enough distance Diagram SP.6 b)	2nd warning board placed at least 45 m (≈50 yds) beyond 1st warning board. 2nd portable AWS magnet placed immediately beyond 1st warning board.	Same minimum spacing rule as consecutive TSRs.

3 section



One TSR inside another

Diagram SP.6 a) **Diagram SP.6 b)**

Diagram SP.6 a) Lower inside higher | SP.6 b) Insufficient distance — GERT8000-SP

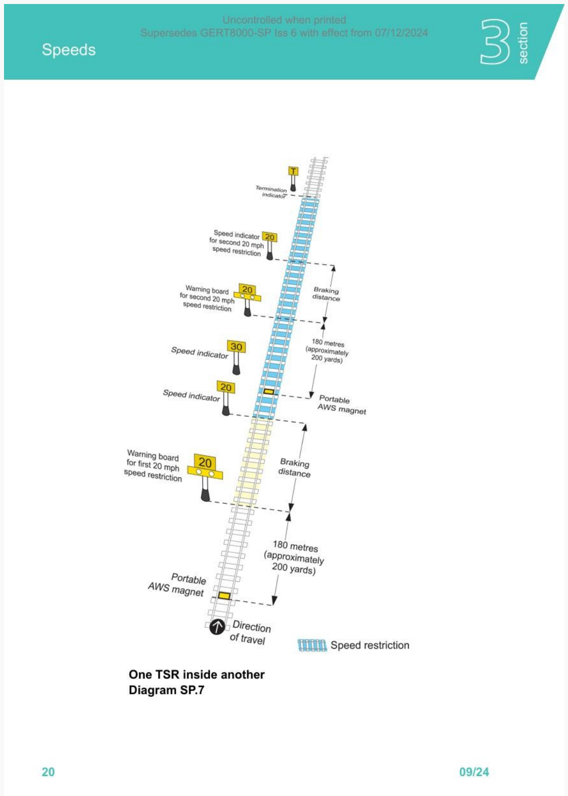


Diagram SP.7 — Higher speed inside lower outer TSR — GERT8000-SP

Memory rule — consecutive and nested TSRs
Whenever two speeds meet: the termination indicator is never placed at an intermediate transition — always a new speed indicator is used instead. The single termination indicator goes at the very end of the whole arrangement. A warning board is only provided when you're about to enter a LOWER speed than you're currently travelling.

7. TSRs at a Diverging Junction (SP §3.8)

a) TSR on diverging route only

- Equipment is normal except the warning board and speed indicator both carry a direction indicator showing the TSR applies to the diverging route only.

b) TSR on one route commences beyond the junction

- Warning boards for both TSRs are positioned on the approach to the junction.
- Only the speed indicator on the diverging route is positioned beyond the junction.
- One warning board has a direction indicator for the diverging route.

c) TSRs on both routes commencing beyond the junction

- Warning boards for both TSRs are on the approach to the junction; speed indicators for both routes are beyond it.
- One warning board has a direction indicator for the diverging route.

- If insufficient distance: the 2nd warning board is at least 45 m beyond the 1st; the 2nd AWS magnet is immediately beyond the 1st warning board.

8. TSR Beyond a Station or Siding Connection (SP §3.9)

This applies where the warning board is on the approach to a passenger station, siding connection, or dead-end platform line.

- If the speed indicator is more than 300 m (≈330 yds) beyond the station/siding connection — a repeating warning board is placed as a reminder.

The repeating warning board is placed at one of:

  - Next to the platform starting signal (if one exists)
  - Next to the siding exit signal
  - Immediately ahead of the station, siding connection, or dead-end platform line

- A portable AWS magnet is NOT provided on the approach to the repeating warning board.

9. TSR Where Trains Can Reverse or Change Drivers (SP §3.10)

- An additional speed indicator is placed within the TSR as a reminder.

Placed at one of:

  - Next to the starting signal
  - Immediately ahead of the station

10. TSR Administration (SP §3.12 – 3.14)

Situation	Rule
Moving a TSR (§3.12)	Can be moved if published in the WON AND: all three boards moved in direction of travel, OR warning board + speed indicator moved toward termination indicator, OR termination indicator moved toward speed indicator.
TSR not introduced (§3.13)	Published at least 24 hours before start in a special notice. Boards are not provided. If less than 24 hours notice: TSR is set up but normal permissible speed applies. Boards/indicator show permissible speed or a SPATE indicator.
TSR eased early (§3.14)	Warning boards and speed indicator are changed to show the higher speed.
TSR removed early (§3.14)	Warning boards and speed indicator show either the permissible speed or a SPATE indicator.

PART 2 — Emergency Speed Restrictions (ESR)

11. Emergency Speed Restriction (SP §4)

An ESR is imposed when an immediate speed restriction is needed before the normal TSR equipment can be put in place.

Before equipment is in place — Signaller (§4.1)

- Stop each train that will travel over the ESR.
- Tell the driver: the location where the ESR begins and ends, and the speed limit imposed.
- Continue these arrangements until the equipment has been set up.

Before equipment is in place — Driver (§4.2)

- Control your train to travel over the affected line at no more than the speed told to you by the signaller.
- Do not increase speed until the whole of your train has passed clear of the lower-speed section.

After equipment has been provided — Driver (§4.2)

- Control speed to no more than the speed shown on the warning board.
- Where differential speeds are shown, observe the speed that applies to your train.

> Key difference between TSR and ESR: With an ESR you may receive your restriction verbally from the signaller before any lineside equipment is in place. With a TSR, equipment is always provided before the restriction comes into force.

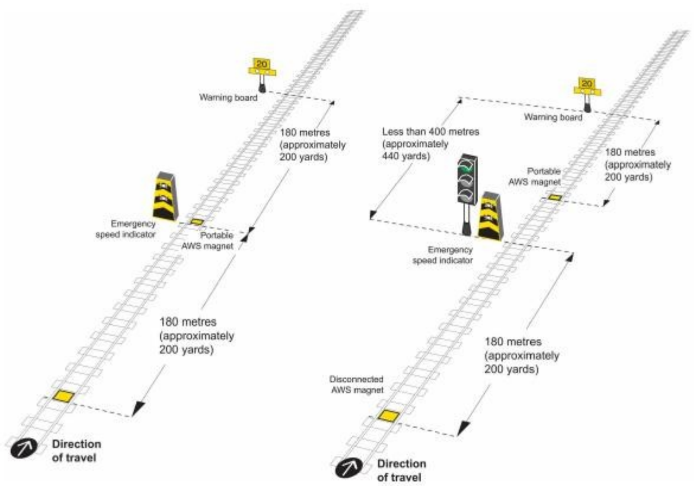
ESR Normal Arrangements (§4.3) — Emergency Indicator

When an ESR is to last for more than a short time, equipment is provided as soon as possible. The normal TSR equipment (§3.2) is provided plus an additional emergency indicator.

Equipment	Distance	Notes
Portable AWS magnet (for emergency indicator)	Normally 180 m (≈200 yds) before emergency indicator	
Emergency indicator	At least 180 m AND not more than 400 m (≈440 yds) before warning board	Must remain until ESR in WON or restriction withdrawn
AWS magnet (for warning board)	At or beyond the emergency indicator	
Warning board	Normal braking	As per standard TSR

Equipment	Distance	Notes
	distance before speed indicator	
Speed indicator	At start of restriction	As per standard TSR
Termination indicator	At end of restriction	As per standard TSR

signaller



Emergency indicator

Diagram SP.14 a) **Diagram SP.14 b)**

Diagram SP.14 a) ESR — normal | SP.14 b) ESR — where fixed AWS magnet exists — GERT8000-SP

ESR Where There is a Fixed AWS Magnet (§4.4)

- Emergency indicator must NOT be placed between a fixed AWS magnet and the equipment it applies to.
- AWS magnet and warning board kept at normal distance if possible; may be reduced to not less than 45 m (≈50 yds).
- Emergency indicator may be placed at the signal — electro-magnet disconnected, no portable AWS magnet provided. Driver will always receive an AWS warning regardless of signal aspect.

Emergency Indicator — How Long It Stays (§4.5)

- The emergency indicator remains in position until: details of the restriction appear in the Weekly Operating Notice, OR the restriction is withdrawn.

PART 3 — Defective Equipment & Blanket Restrictions

12. Defective or Missing Equipment (SP §5)

Missing or incorrect boards/indicators (§5.1)

Tell the signaller IMMEDIATELY (stop your train if necessary) if a warning board, repeating warning board, or speed indicator is:

- Missing
- In a different place from the one published in the Weekly Operating Notice
- More restrictive than that shown in the Weekly Operating Notice
- The speed on the DMI differs from lineside equipment or the WON

You do not need to tell the signaller if you have already been told about this.

Boards or indicators becoming difficult to see (§5.2)

- Tell the signaller at the first opportunity if a warning board, repeating warning board, or speed indicator is, or is becoming, difficult to see.

Defective or missing emergency indicator (§5.3)

- Tell the signaller immediately if anything is wrong with the emergency indicator, stopping the train specially if necessary.

Driver duties	Signaller duties
Tell signaller immediately	Report defect to Operations Control
Stop train if necessary	Tell driver of each approaching train until fixed
Report if board is missing, wrong position, or more restrictive than WON	For defective emergency indicator: stop each approaching train and tell driver about ESR
Report if difficult to see at first opportunity	

13. Blanket Speed Restrictions (SP §6)

- Emergency indicators and other track equipment are NOT provided for a blanket speed restriction.
- The signaller (told by Operations Control) arranges for the driver of each affected train to be told: the speed restriction and the locations between which it is to be observed.
- This is not required if Operations Control has arranged to tell drivers by other means.

Revision Q&A

All answers are sourced from GERT8000-SP.

Q1: What is the normal distance between the portable AWS magnet and the warning board for a TSR?

A: Normally 180 metres (approximately 200 yards). This may be reduced to not less than 45 metres (approximately 50 yards) in certain circumstances.

Q2: What is the normal distance between the warning board and the speed indicator?

A: The appropriate braking distance for the permissible speed at that location.

Q3: Where is the termination indicator placed in a standard TSR?

A: At the end of the TSR. An acceleration indicator may be used as an alternative in some circumstances.

Q4: When can you increase speed after passing through a TSR?

A: Only after the whole of your train has passed clear of the section of line with the lower speed.

Q5: If there are two consecutive TSRs with a lower speed beyond a higher speed, what happens to the termination indicator at the end of the first (higher speed) TSR?

A: It is not provided. Instead, a speed indicator showing the speed for the lower speed TSR is placed there.

Q6: If there are two consecutive TSRs with a higher speed beyond a lower speed, what equipment is NOT provided for the second TSR?

A: The warning board is not provided for the second (higher speed) TSR.

Q7: With consecutive TSRs, where is the single termination indicator located?

A: At the end of the second TSR.

Q8: How close can the second warning board be placed to the first when there is insufficient distance between consecutive or nested TSRs?

A: At least 45 metres (approximately 50 yards) beyond the first warning board.

Q9: If a lower speed TSR sits inside a higher speed TSR, what replaces the termination indicator at the end of the inner (lower speed) TSR?

A: A speed indicator showing the speed of the outer (higher speed) TSR.

Q10: Where is the warning board placed if a TSR is at a signal with a fixed AWS magnet — and what happens to the electro-magnet?

A: The warning board is placed at the signal. The associated electro-magnet is disconnected and no temporary AWS magnet is provided. The driver will always receive an AWS warning indication regardless of the signal aspect.

Q11: On a single or bi-directional line, what additional equipment is provided for a TSR?

A: Equipment is provided in both directions. A cancelling indicator is placed normally 180 metres (approx. 200 yards) beyond the AWS magnet at each end, facing trains that have already passed through the restriction. May be reduced to 45 metres minimum.

Q12: When must a repeating warning board be provided for a TSR beyond a station?

A: When the speed indicator is more than 300 metres (approximately 330 yards) beyond the station or siding connection.

Q13: What additional equipment is provided where a TSR is at a location where trains can reverse or regularly change drivers?

A: An additional speed indicator is placed within the TSR as a reminder, next to the starting signal or immediately ahead of the station.

Q14: What three options exist for moving a published TSR?

A: (1) All three boards moved in direction of travel; (2) Warning board and speed indicator moved toward the termination indicator; (3) Termination indicator moved toward the speed indicator.

Q15: What is the key difference between a TSR and an ESR from the driver's perspective?

A: With an ESR, the driver may receive the restriction verbally from the signaller before any lineside equipment is in place. With a TSR, equipment is always set up before the restriction comes into force.

Q16: What must the signaller tell the driver before an ESR is equipped?

A: The location where the ESR begins and ends, and the speed limit imposed.

Q17: What is the minimum and maximum distance the emergency indicator must be placed before the warning board?

A: At least 180 metres (approx. 200 yards) and not more than 400 metres (approx. 440 yards) before the warning board.

Q18: When does the emergency indicator come out of use?

A: When details of the speed restriction appear in the Weekly Operating Notice, or when the restriction is withdrawn.

Q19: What must you do if you see a warning board that is more restrictive than the speed shown in the Weekly Operating Notice?

A: Tell the signaller immediately. Stop your train specially if necessary. You do not need to report it if you have already been told about it.

Q20: What equipment is NOT provided for a blanket speed restriction?

A: Emergency indicators and other track equipment are not provided. Drivers are told about the restriction and its locations by the signaller (or by other means arranged by Operations Control).

End of TSR & ESR Revision Guide | GERT8000-SP